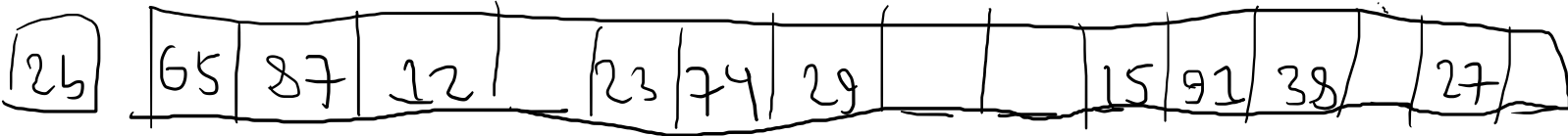
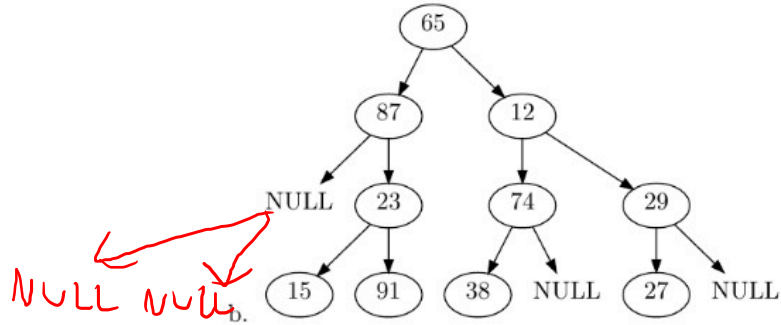
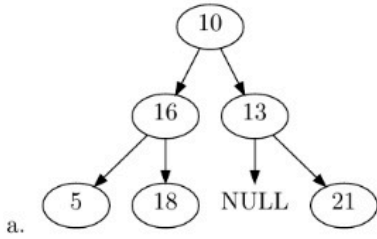


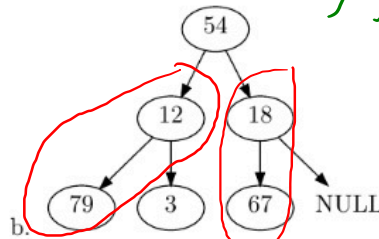
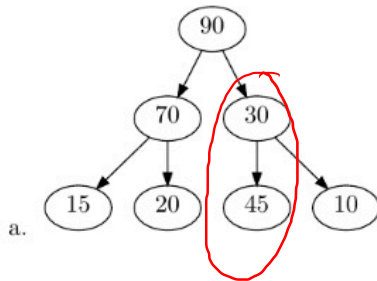
2. Representasi array dari binary tree

Untuk masing-masing binary tree berikut ini, tuliskan representasi array nya.



3. Hampir Max Heap

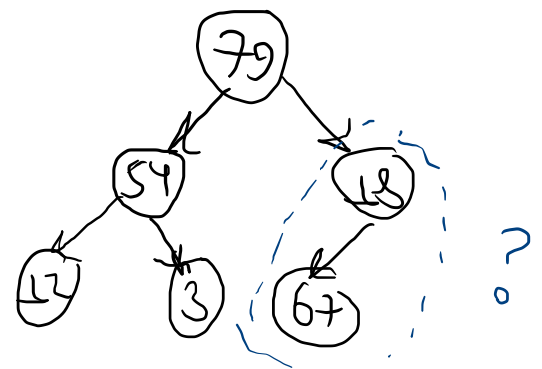
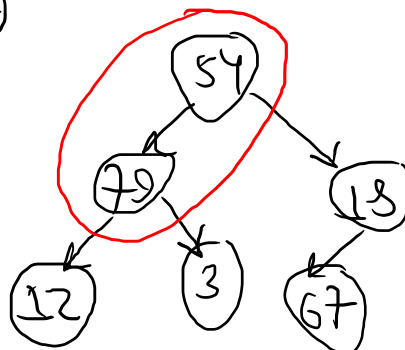
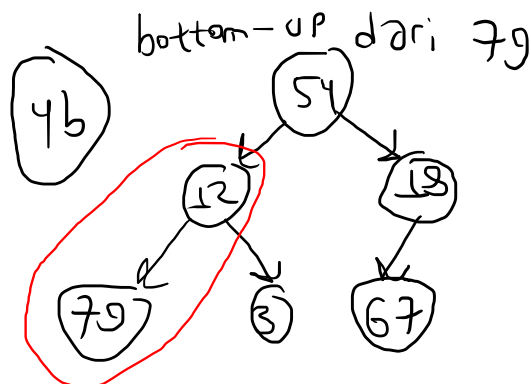
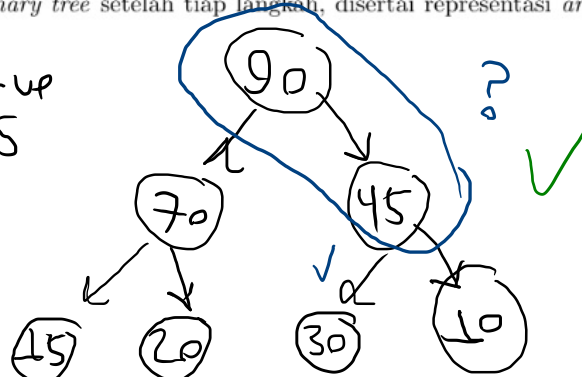
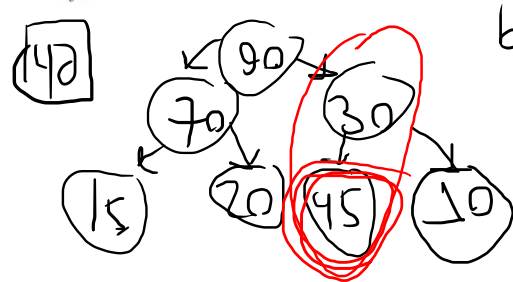
Suatu max heap adalah binary tree yang bersifat complete dan memenuhi max heap property, di mana data pada suatu node selalu lebih besar atau sama dengan data pada semua child nya. Untuk masing-masing binary tree berikut ini, periksalah apakah tiap node sudah memenuhi max heap property. Lingkari semua pasangan node dengan child nya yang belum memenuhi max heap property, kalau ada.

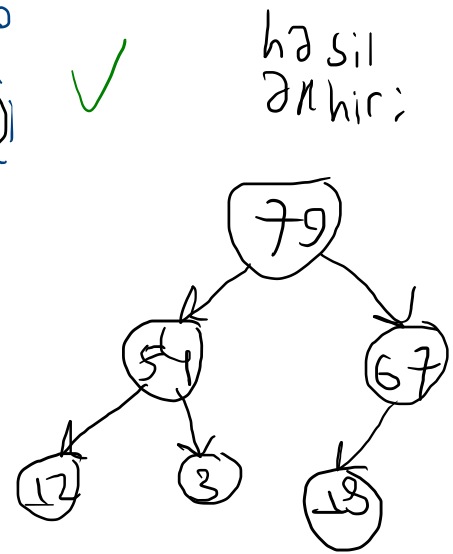
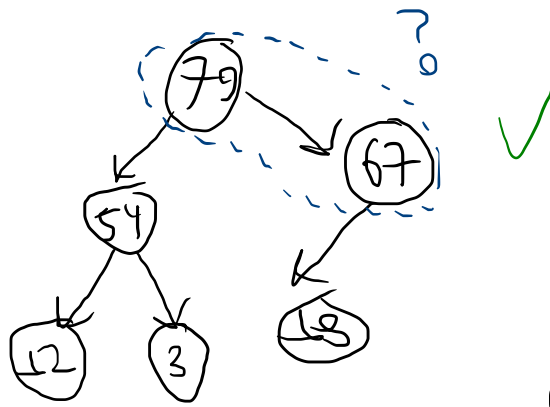
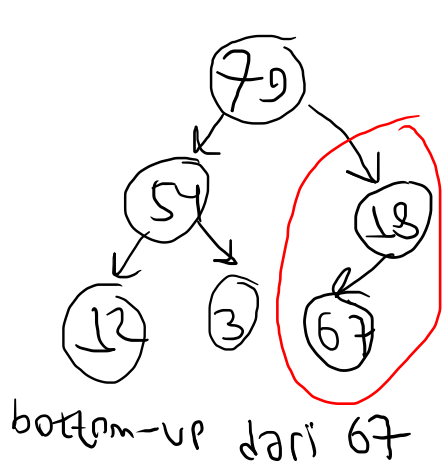


$$A[\text{parent}] \geq A[\text{child}]$$

4. (Max Heap) heapify: bottom-up

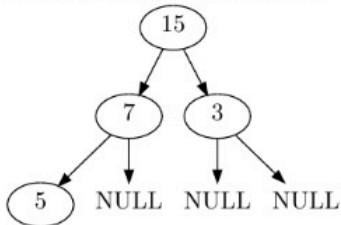
Untuk tiap binary tree di soal no. 3, lakukan heapify (bottom-up), dimulai dari node yang sesuai, agar binary tree berubah menjadi max heap. Gambar keadaan binary tree setelah tiap langkah, disertai representasi array nya.





5. (Max Heap) Insertion

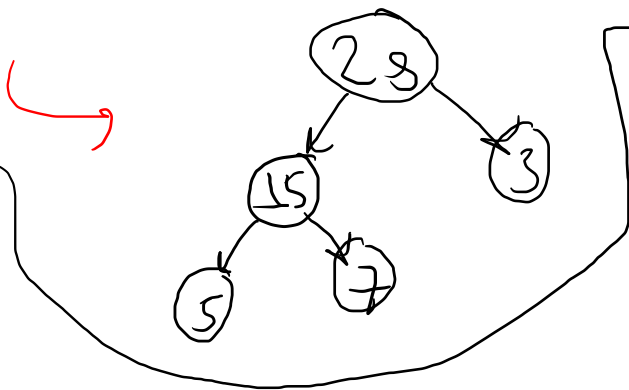
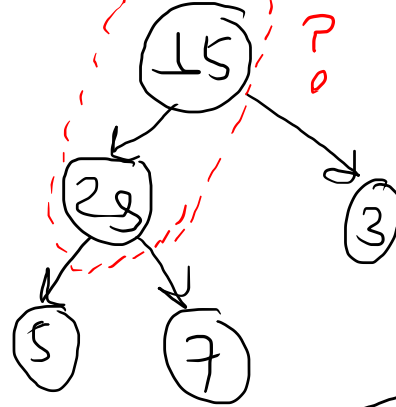
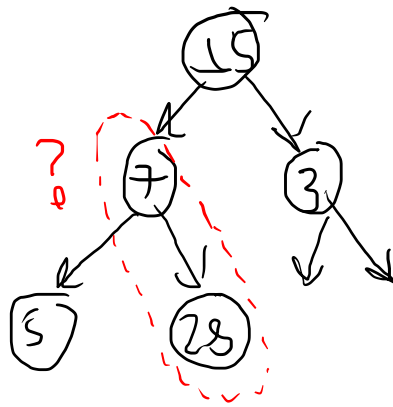
Perhatikan max heap berikut.



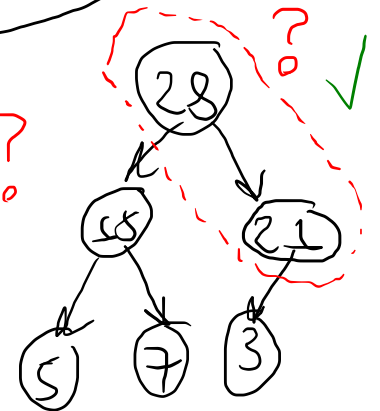
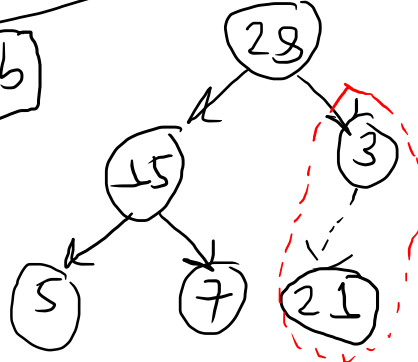
Lakukan *insertion*, satu-per-satu secara bertahap, tiap nilai berikut ke *max heap*, hingga pada akhirnya *max heap* secara keseluruhan terdiri dari 7 node. Apabila *insertion* melibatkan *heapify*, gambarkan keadaan *max heap* di tiap tahap *heapify*.

- $x = 28$
- $x = 21$
- $x = 35$

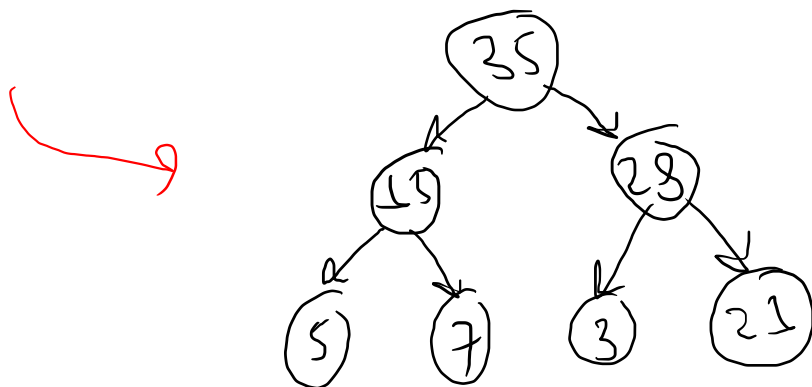
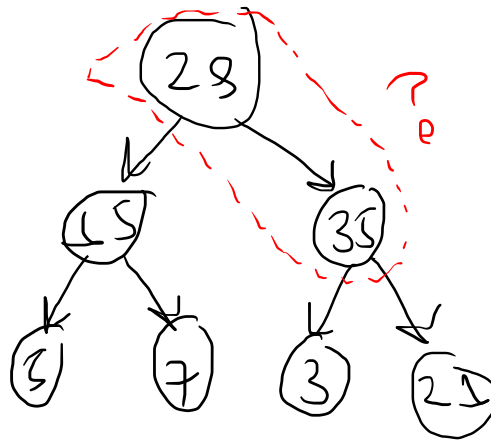
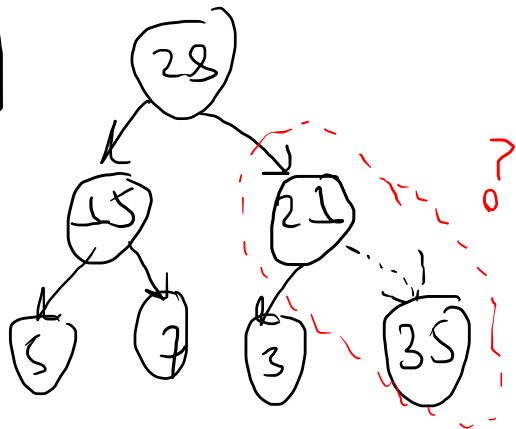
Sa



Sb



5c



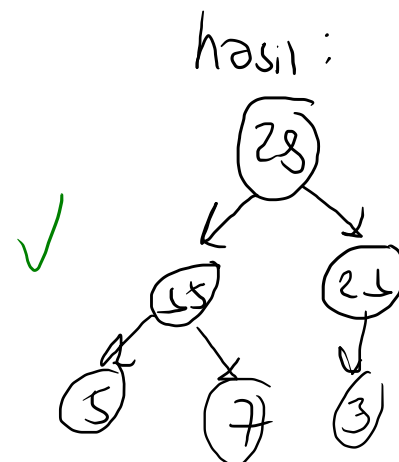
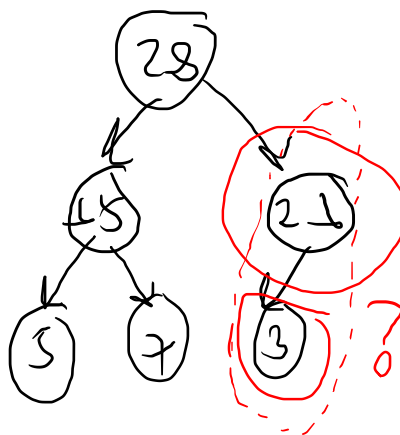
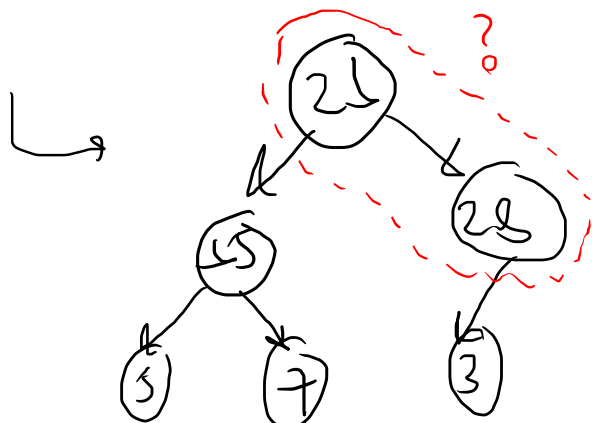
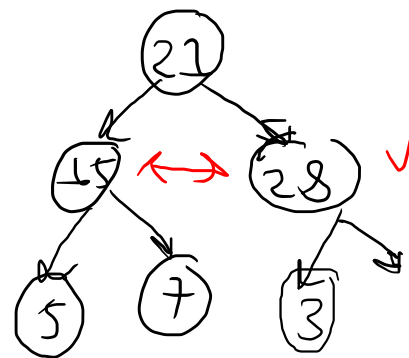
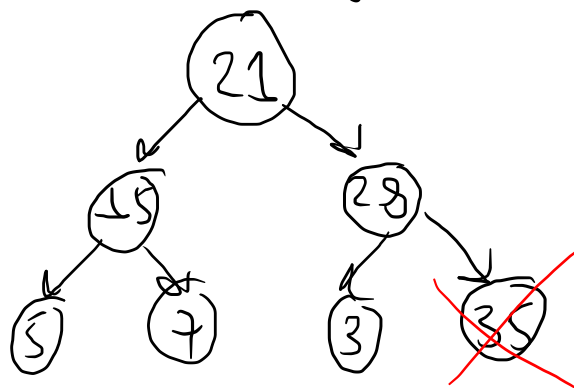
6. (Max Heap) Deletion dan heapify: top-down

Pada max heap hasil mengerjakan soal no. 5, lakukan deletion berkali-kali hingga max heap hanya terdiri dari

- a. 6 node
- b. 5 node
- c. 4 node
- d. 3 node

deletion di heap → hapus root

6d



hasil :